

"Periodic Classification of Elements"

⇒ Periodic Table: The arrangement of elements on the basis of Periodic law in the form of a table is known as Periodic table.

⇒ Dobereiner's Triads law:

A German chemist Dobereiner observed relationship between atomic masses of several groups of three elements called triads. In these groups, the central or middle element have atomic mass average of the other two elements. One triad group example is that of Ca (40), Sr (88), and Ba (137). The atomic mass of Strontium is the average of atomic mass of Ca & Ba.

⇒ Newlands Octaves Law:

British Chemist Newlands noted that there was a repetition in chemical properties of every eighth element if they were arranged by their increasing atomic masses.

⇒ Mendeleev's Periodic Table:

Russian Chemist, Mendeleev arranged the known elements (only 63) in order of increasing atomic masses, in horizontal rows called periods and elements with similar properties were in the same vertical columns.

This arrangement of elements was called periodic table.

Mendeleev states the Periodic law.
as "properties of the elements are
periodic functions of their atomic masses."

But taking atomic masses as the basis
of periodic table some elements position
aren't suit to other elements of
their respective columns.

Examples of elements pair, whose position
in the P.T. can't be explained by the
Mendeleev's Periodic law.

Such as Ar (39.9), K (39.1) &
I (127.6), Te (126.9).

⇒ Modern Periodic Law:

In 1913. H. Moseley discovered a new property of the elements i.e. atomic number.

The amended modern periodic law was states as, " Properties of the elements are periodic function of their atomic numbers."

☐ Salient Features of Long Form of Periodic Table:

- i. This table consists of seven horizontal rows called periods.
- ii. First period consists of only two elements. Second and third

periods consist of 8 elements each. Fourth and fifth periods consist of 18 elements each. Sixth and Seventh periods, has 32 elements.

iii. Elements of a period show different properties.

iv. The elements of a group show similar chemical properties.

v. Elements are classified into 4 blocks depending upon the type of the subshell which gets the last electron. These blocks are s, p, d and f.

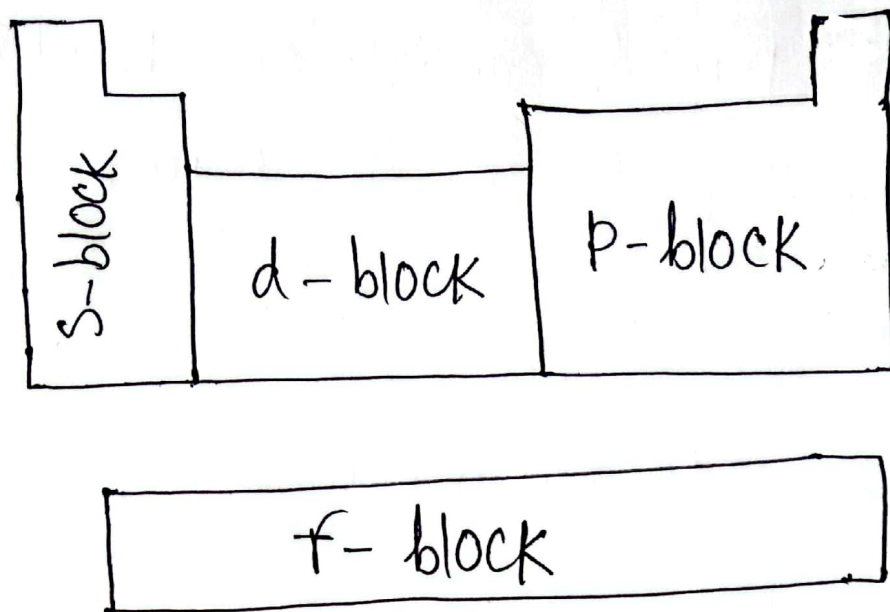
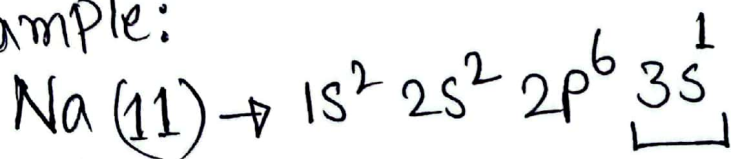


Fig: Modern Periodic table
Showing four blocks.

⇒ Position of elements in periodic table:

1. For s-block elements, electron exist in the last subshell (s-orbital) indicates their group number and highest number of 'n' indicates their period number.

Example:



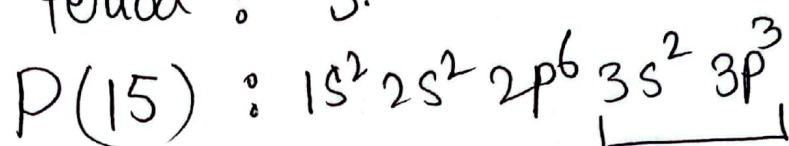
Group: 1

Period: 3.

2. For p-block elements, the total electron exist in last s and p-orbital indicates their group number. Adding 10 with total electrons of s and p-orbitals, we can find their gr.
e.g., $\text{Al}(13) \rightarrow 1s^2 2s^2 2p^6 \underline{3s^2 3p^1}$

$$\text{Group: } 2 + 1 + 10 = 13$$

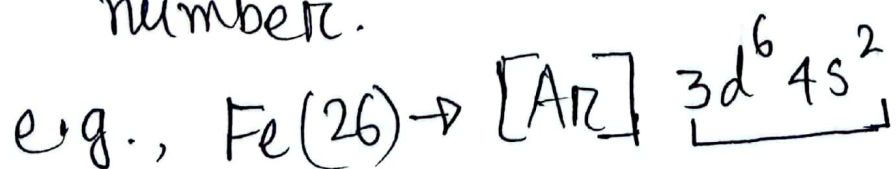
$$\text{Period: } 3$$



$$\text{Group: } 2 + 3 + 10 = 15$$

$$\text{Period: } 3$$

3. For d-block elements, the total number of electron exist in $(n-1)d$ and ns-orbital indicates their group number.



$$\text{Group: } 6 + 2 = 8$$

$$\text{Period: } 4$$

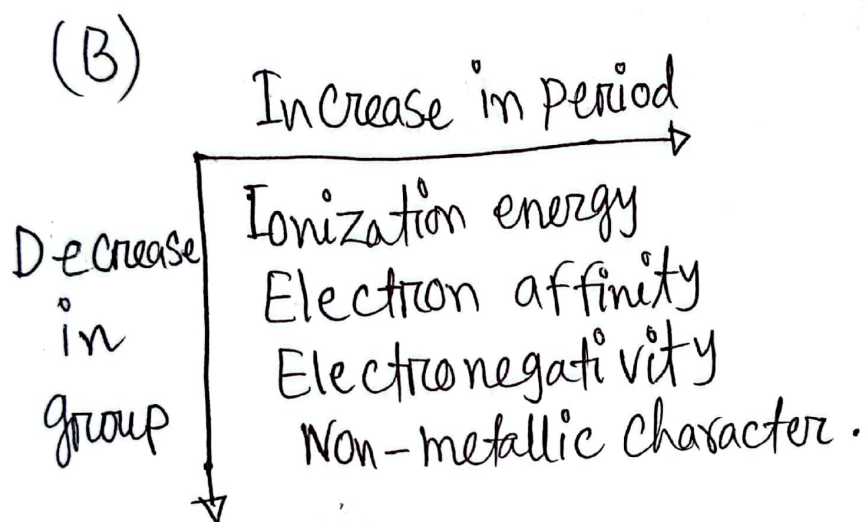
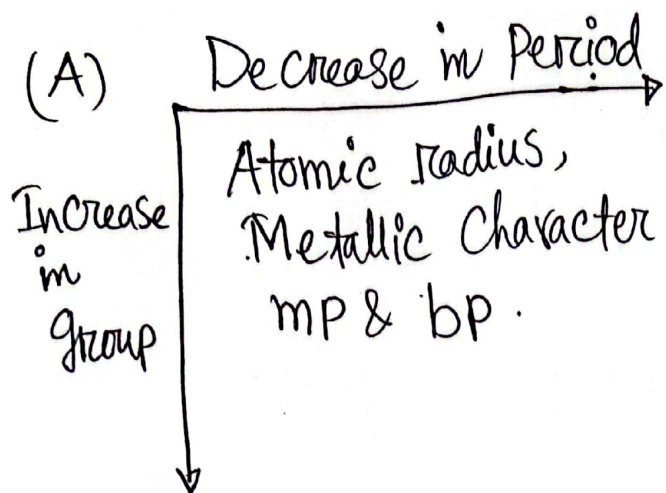
4. f-block elements are exist in group 3 (three) and in 6th and 7th periods.

⇒ Periodicity of Properties:

The properties which comes after definite intervals of atomic number (i.e., 2, 8, 8, 18, 18, 32 etc). are called Periodicity of Properties or Periodic Properties.

⇒ Periodic Properties:

1. Atomic Radius
2. Ionic Radius
3. Ionization energy
4. Electron affinity
5. Electronegativity
6. Metallic and Non-metallic Character
7. Oxidation states etc.



1. Atomic Radius:

In Periods atomic radius of the elements decrease periodically and in groups atomic radius increase periodically.

In Periods one electron increase in the outermost shell and one proton increase in the nucleus. The attraction between Proton (Nucleus) and electron increase

Periodically. Thus their atomic radius decrease in Periods.

On the other hand, in groups one energy level is increase in every case and this happens Periodically. Thus their atomic radius increase in groups.

⇒ The less atomic radius the more atoms are binding together.

For this reason mp, bp, IP, EA and Electronegativity increase in the Periods.

⇒ The more atomic radius, the less atoms are binding (weakly binding) together. For this reason their mp, bp, IP, EA and Electronegativity decrease in the groups.

⇒ 2. Ionic Radius:

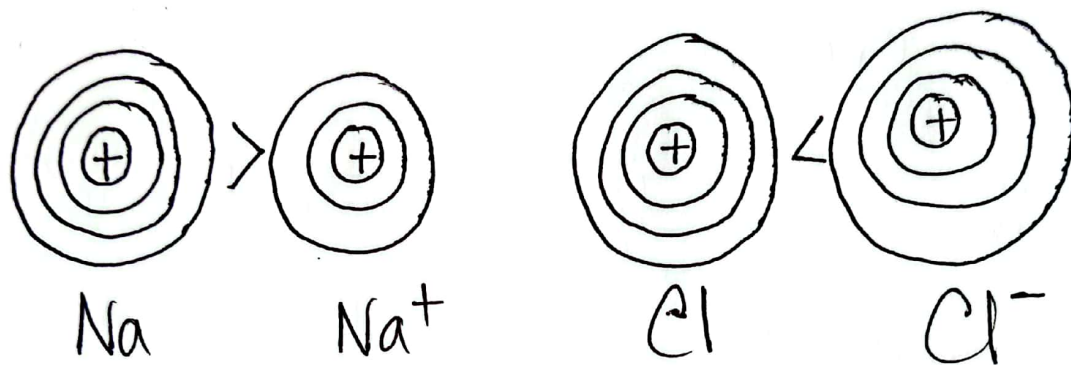
The removal of an electron from an atom results in the formation of a cation, whereas gain of an electron leads to an anion.

→ The ionic radii can be estimated by measuring the distances betⁿ cations and anions in ionic crystals.

→ A cation is smaller than its parent atom because it has fewer electrons while its nuclear charge remains unchanged.

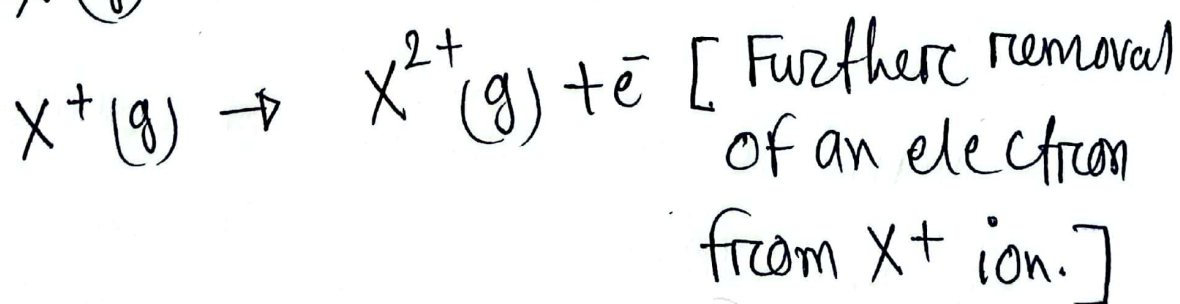
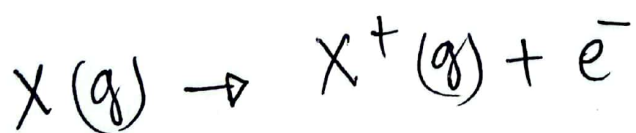
→ The size of an anion will be larger than that of the parent atom because the addition of one or more electrons would result in increased repulsion among the electrons and a decrease in effective nuclear charge.

For example, the ionic radius of Na (i.e. Na^+) and Na atom and ionic radius of Cl (i.e. Cl^-) and Cl atom are shown below.



3. Ionization Energy:

The energy required to remove an electron from an isolated gaseous atom (X) in its ground state is known as ionization energy or ionization potential.



Ionization energy is increase in Periods and decrease in groups.

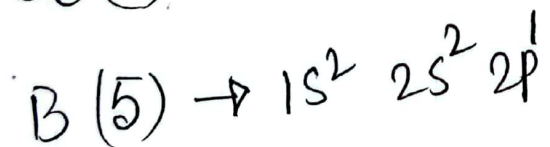
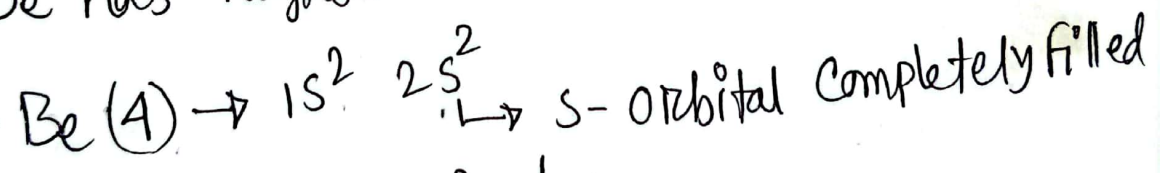
Ionization energy of elements of Period 2 is shown below:

Element	Li	Be	B	C	N	O	F	Ne
Ip	520	899	801	1086	1402	1314	1681	2080

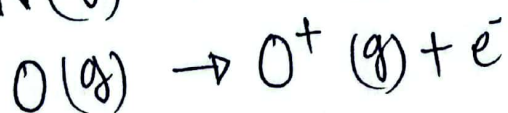
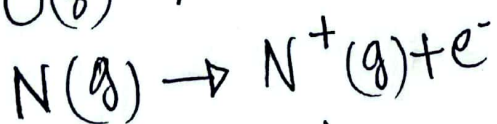
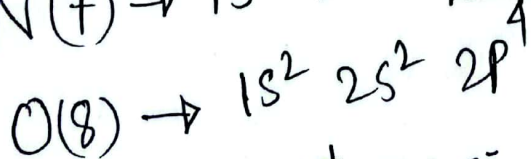
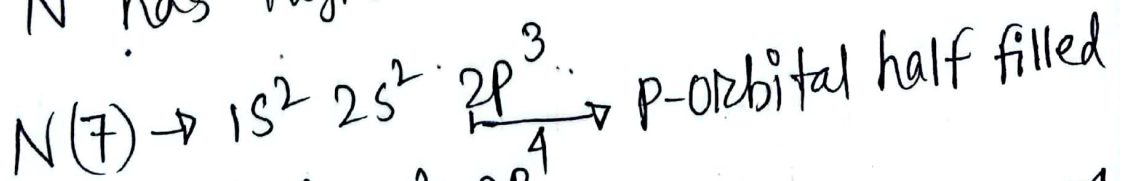
(Ip shown kJ mol^{-1} unit)

⇒ Anomalous Ip in Period 2 elements:

i. Be has higher Ip than B.



ii. N has higher Ip than O.



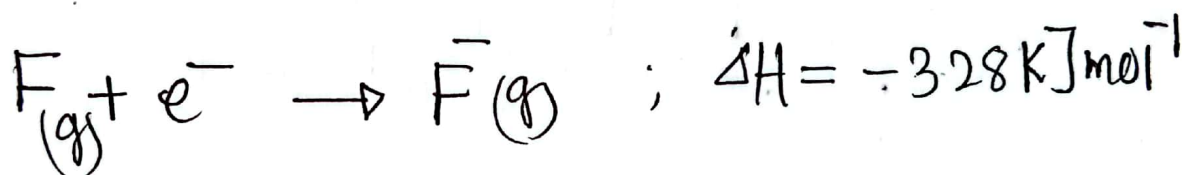
$$\Delta H = +1402 \text{ kJ mol}^{-1}$$

$$\Delta H = +1314 \text{ kJ mol}^{-1}$$

4. Electron Affinity (EA):

Electron affinity is defined as the amount of energy released when an electron is added to the outermost shell of an isolated gaseous atom.

Example:



→ EA of elements increases in periods and decreases in groups.

⇒

2nd period elements	Li	Be	B	C	N	O	F	Ne
EA (kJ mol ⁻¹)	-60	>0	-29	-122	0	-141	-328	0

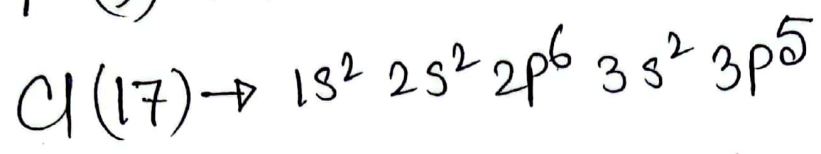
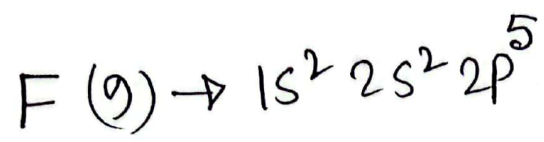
→ EA increases in period.

⇒

17th gr elements	F	Cl	Br	I
EA (kJ mol ⁻¹)	-328	-349	-325	-295

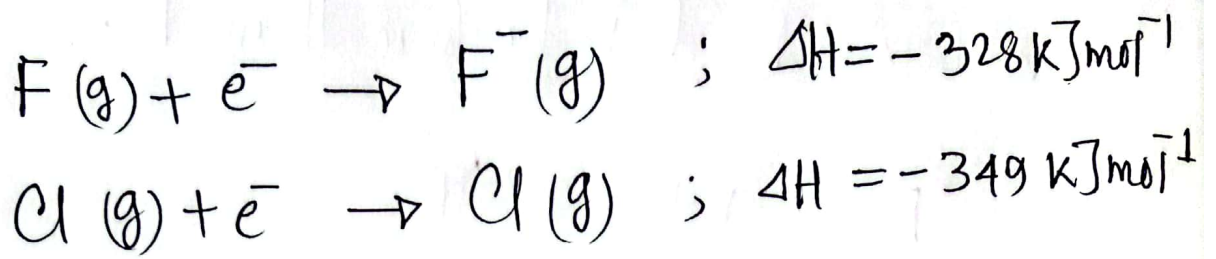
→ EA decreases in group

⇒ EA of Cl is higher than F.



Both F and Cl atom has 7 electrons in their outermost shell. But F has only two energy levels and exist 7 electrons in its outermost shell. Repulsion among electrons in outermost shell is greater because of its small atomic radius. Hence, addition of extra electron isn't welcomed here properly.

On the other hand Cl atom has three energy levels and exist 7 electrons of the outermost shell in large atomic radius of Cl. Hence, repulsion among electrons is less and show greater electron affinity.



5. Electronegativity:

The ability of an atom to attract the shared pair of electron towards itself in a molecule, is called electronegativity.

→ Electronegativity value increase in Period from left to right and decrease in group from up to down.

⇒

2nd Period elements	Li	Be	B	C	N	O	F
EN	1.0	1.5	2.0	2.5	3.0	3.5	4.0

↳ Increase in Period

⇒

Elements of gr 17	F	Cl	Br	I
EN	4.0	3.2	3.0	2.7

↳ Decrease in group.

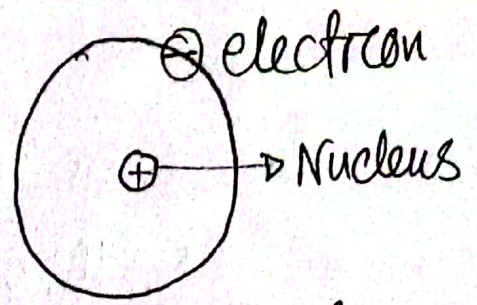
⇒ Rutherford's Atomic Model:

Ernest Rutherford, a British Scientist conducted an experiment and based on the observations of this experiment, he explained the atomic structure of elements and proposed Rutherford's Atomic Model.

⇒ Postulates of Rutherford's Model:

1. Atom has a tiny dense central core which contains practically the entire mass and positive charge of the atom. This central region is called nucleus and rest of the part of atom almost empty.
2. The negatively charged electrons are moving in orbits or closed circular paths around the nucleus like planets around the sun.

3. Electrons being negatively charged and nucleus being a densely concentrated mass of positively charged particles are held together by a strong electrostatic force of attraction.



⇒ Limitations of Rutherford's Model:

1. According to Maxwell's theory, any charged particles with acceleration moving round the nucleus would continuously emit electromagnetic radiation and lose energy. And continuously decrease its path lengths and eventually fall into the nucleus and the atom would collapse. So, according to Maxwell's theory it can't explain the stability of an atom.

2. One of the drawbacks of the Rutherford model was also that he did not say anything about the arrangement of electrons in an atom.
 3. It can't explain or didn't say anything about line spectrum.
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Bohr's Atomic Model

In 1913 Niels Bohr proposed a new model of the atom which explained some of these things (Orbit, Absorption, emission) and also the emission spectrum of hydrogen.

⇒ Postulates of Bohr's Model:

1. Electrons travel around the nucleus in specific circular orbits.
Electrons in each orbit have a definite

energy and are at a fixed distance from the nucleus. The orbits are given the letter designation n and each is numbered 1, 2, 3 etc (or, K, L, M, etc) as the distance increases from the nucleus.

2. While in these specific orbits, an electron does not radiate or absorb energy. Hence the specific

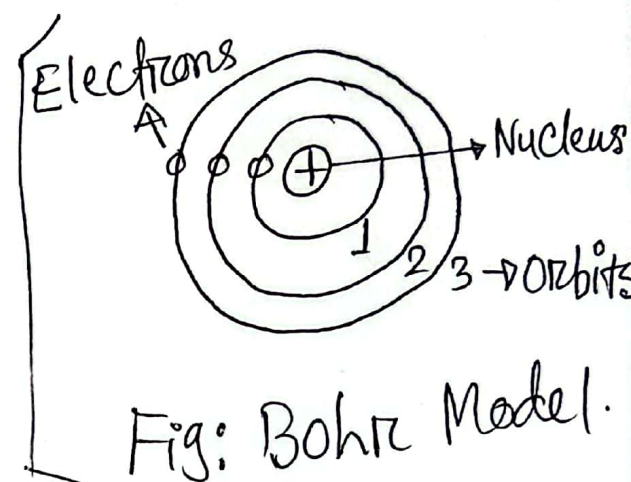


Fig: Bohr Model.

orbits available to the electron in an atom are referred to as 'stationary energy levels' or simply energy levels.

3. An electron can move from one energy level to another by quantum or photon jumps only (or, by absorption or emission

of energy). The quantum or photon energy absorbed or emitted is the difference betⁿ

the lower and higher energy levels of the atom.

$$\text{i.e., } \Delta E = E_{\text{high}} - E_{\text{low}} = h\nu$$

Where, h = Planck's constant
 $= 6.626 \times 10^{-34} \text{ Js}$

ν = Frequency of a photon emitted or absorbed energy.

4. The angular momentum (mvr) of an electron orbiting around the nucleus is an integral multiple of Planck's constant divided by 2π .

$$\text{So, Angular momentum} = mvr = n \frac{h}{2\pi}$$

Where, m = mass of electron,

v = Velocity of the electron,

r = Radius of the orbit,

n = 1, 2, 3 etc, &

h = Planck's Constant.

⇒ Limitations of Bohr Model:

1. It violates the Heisenberg Uncertainty Principle. The Bohr atomic model theory considers electrons to have both a known radius and orbit i.e. known position and momentum at the same time, which is impossible according to Heisenberg.
2. The Bohr atomic model theory made correct predictions for smaller sized atoms like hydrogen, but spectral predictions are obtained when larger atoms are considered.
3. It failed to explain the Zeeman effect when the spectral line is split into several components in the presence of a magnetic field.
4. It failed to explain the Stark effect when the spectral line gets split up into fine lines in the presence of electric field.

⇒ Success of Bohr Model:

1. The main achievement of Bohr's theory of atom is it can explain the stability of the atom.
2. It can successfully explain the line spectrum of hydrogen and hydrogen like single electron containing ions like He^+ and Li^{2+} ions spectrum.
3. Through Bohr theory can predict correct size of Hydrogen atom.
4. Through Bohr theory can predict the frequencies in the Hydrogen emission spectrum.

Quantum Numbers:

The set of numbers used to describe the position and energy of electron in an atom are called quantum numbers.

There are four quantum numbers. They are-

1. Principal quantum number (denoted by n)
2. Orbital angular momentum quantum number (Or, azimuthal quantum number) denoted by l .
3. Magnetic quantum number (denoted by m_l)
4. The Electron spin quantum number (denoted by m_s or s)

⇒ Four Quantum Numbers:

Number	Symbol	Possible Values
Principal QN	n	1, 2, 3, 4...
Angular momentum QN	l	0, 1, 2, 3... ($n-1$)
Magnetic QN	m_l or m	$-l, \dots, -1, 0, 1, \dots, +l$
Spin QN	m_s or s	$+\frac{1}{2}, -\frac{1}{2}$

1. Principal Quantum Number (n):

- The quantum number which describes the size of the orbital is known as principal quantum number.
- It gives the energy of electron in an orbit.
- It gives idea about shell in which the electron is found.
- The possible values of n are 1, 2, 3 etc.

When,

$n=1$, the electron is in K shell

$n=2$, the electron is in L shell

$n=3$, the electron is in M shell.

2. Azimuthal quantum number (l):

- The quantum number which describes the shape of the orbital is known as azimuthal quantum number.

→ It gives idea about sub-shell in which the electron is located.

→ It also gives the orbital angular momentum of the electron.

→ For a given value of n , the possible values are $l = 0, 1, 2, \dots (n-1)$.

For, $n=1$ $l=0$

$n=2$ $l=0$ and 1

$n=3$ $l=0, 1$ & 2 .

$l=0$, Represents s-orbital

$l=1$, Represents p-orbital

$l=2$, Represents d-orbital

$l=3$, Represents f-orbital.

3. Magnetic Quantum Number (m or m_s):

- It gives information about the Orientation of Orbitals in Space.
- For a given 'l' value, there are $2l+1$ possible values for m .
- These values are given by $m = -l$ to 0 to $+l$.

For, $l = 0$, $m = 0$, s sub-shell contains only one orbital.

For, $l = 1$, $m = -1, 0, +1$

p-subshell contains three orbitals.

For, $l = 2$, $m = -2, -1, 0, +1, +2$.

d-subshell contains five orbitals called d-orbitals (d_{xy} , d_{yz} , d_{zx} , $d_{x^2-y^2}$, d_{z^2})

4. Spin Quantum Number (s):

- It gives information about the direction of spin of electrons.
- The values for s may be $+\frac{1}{2}$ or $-\frac{1}{2}$
- $+\frac{1}{2}$ represents clock-wise spin.
- $-\frac{1}{2}$ represents anticlock-wise spin.

⇒ Significance of four quantum numbers:

1. From Principal quantum number we can get idea about size of the orbit or energy level & For H atom & one electron containing ion we can determine radius of the energy level, velocity of the electron and energy of electron exist in n -th shell.
2. From azimuthal quantum number we can get information about shape of the orbitals.

Thus, from azimuthal quantum number we can know shape of s-orbital is spherical, shape of the p-orbital is dumbbell shaped and shape of the d-orbital is double-dumbbell shaped.

3. From magnetic quantum number we can get information about the three dimensional orientation of orbitals. We also get information about the splitting of spectral line by magnetic field.

4. From spin quantum number, we can know nature and spin direction of electron.

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